Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

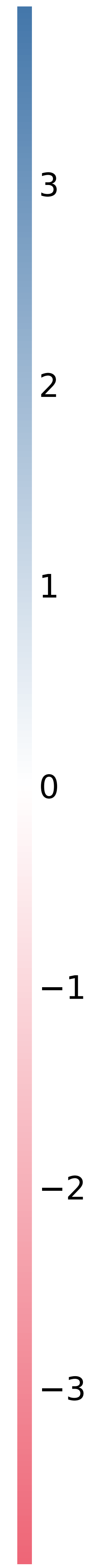
F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal



Reasoning + Answer:

<think>

For part (a):

- The specific heat of water is 1 cal/gÂ°C.

- 1 liter of water weighs approximately 1000 grams.

- We need to find the heat required to increase the temperature from 30Â°C to 60Â°C.

- Using the formula: Q = mass * specific heat * change in temp.

- Q = 1000 g * 1 cal/gÂ°C * (60Â°C - 30Â°C) = 30000 cal = 30 Kcal

For part (b):

- The specific heat of aluminum is 0.215 cal/gÂ°C.

- The mass is 1000 grams (1 kg).

- We need to find the heat required to increase the temperature from 30Â°C to 60Â°C.

- Q = 1000 g * 0.215 cal/gÂ°C * (60Â°C - 30Â°C) = 3225 cal âīĪ 32 Kcal

</think>

<answer>

30 Kcal, 32 Kcal

</answer>